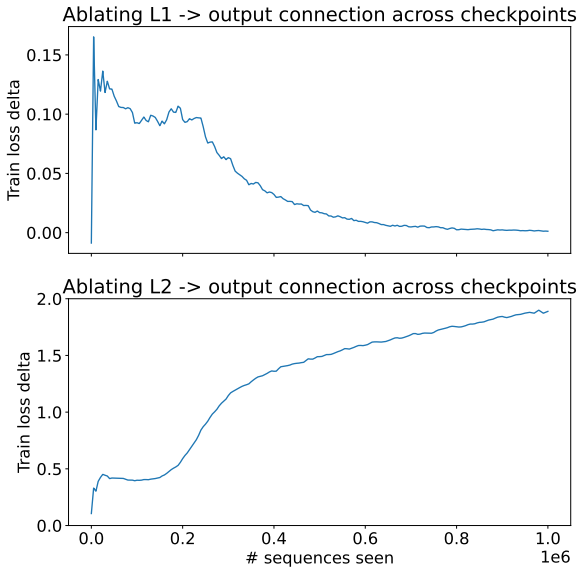

Figure 12. **a)** Change in loss after ablating connection from Layer 1 heads to output (unembedding) layer. Lower change indicates that Layer 1 head to output connections are less important. The ablation was applied independently to each checkpoint from the training run shown in Figure 3 and can thus be viewed as a progress measure. We can see that before the phase change, Layer 1 heads are contributing to the loss, but as induction heads form and the loss goes to 0 at the end of training, Layer 1 heads do not contribute to the output in a direct way. **b)** Same as a), but instead the connection from Layer 2 heads to output (unembedding) layer is ablated. Through training, importance of Layer 2 heads is increasing, especially after the phase change that corresponds to induction head formation.

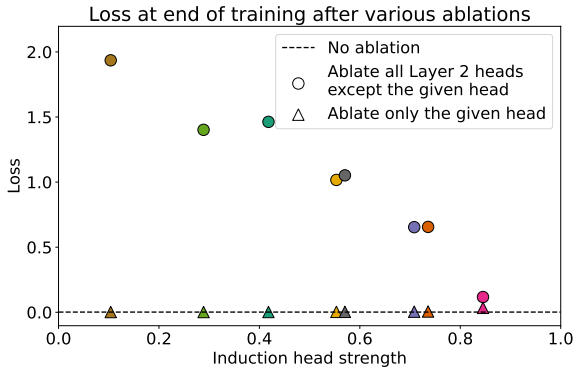

Figure 13. Same as Figure 4a, but plotting loss instead of accuracy. We include this plot for completeness, and used the accuracy version as we felt it was clearer for the main text.

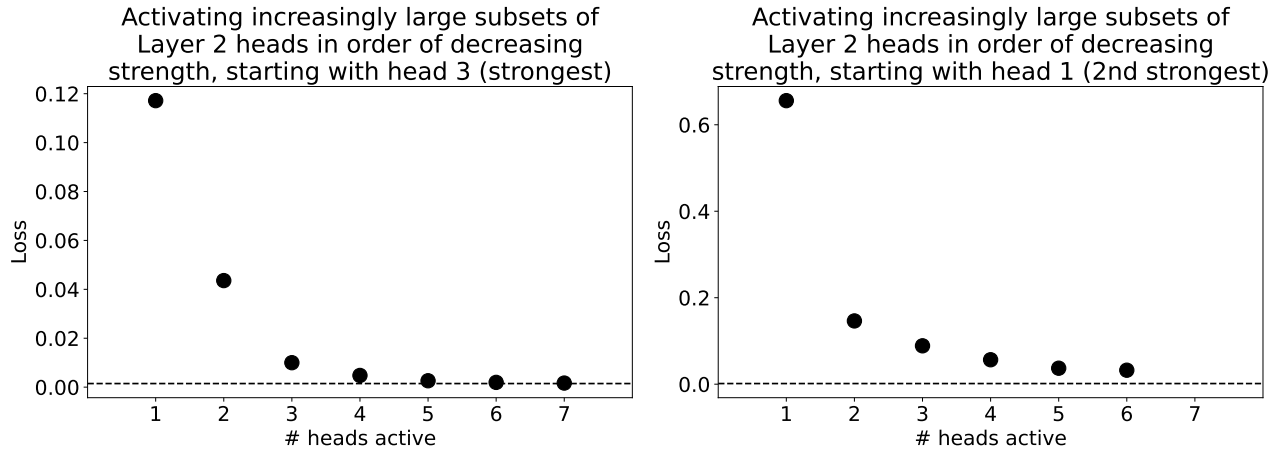

Figure 14. Another way of seeing the additive effect of induction heads. We first order the heads by strength (this happens to be `[3,1,2,7,5,0,4,6]` in our case, see Figure 3b). Then, we consider an ablation where only a certain number of heads (x-axis) are left active. In the left plot, we start with Head 3, so the points from left to right indicate [only Head 3 active], [Heads 3 and 1 active], [Heads 3 and 1 and 2 active], and so on. The right plot is the same, except we never activate Head 3 (the strongest head). The monotonically decreasing loss in both plots indicate that heads have an additive effect on each other. Dotted line indicates loss of full network (no ablations). Note: the y-axis on the right is scaled larger since Head 3 (the strongest head) is never active (so losses are higher).

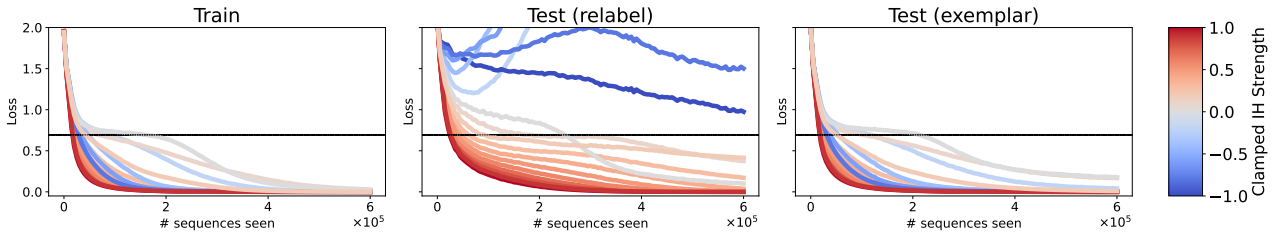

Figure 15. Clamping through training with an imperfect match circuit. For this figure, we clamped the attention pattern of a single Layer 2 head to look like a noisy induction head (the other heads were left active) to see how "strong" an induction head needs to be before it induces the exponential learning dynamics seen when clamping to maximum strength (green line, Figure 5). For example, when "Clamped IH strength" is 0.4, this corresponds to attention of 0.7 to the correct label token and 0.3 to the incorrect label token, and 0 to all other tokens. We vary the noisiness in the induction head and see how this effects dynamics. The darkest red curve (perfect match) corresponds to just learning a copy subcircuit (as in Figure 5, green line), exhibiting a smooth exponential decay pattern. As the strength is turned down to 0, we see the learning dynamics shift from exponential to slow exponential to the usual phase change when the IH strength is clamped to 0 (as this head is ignored, and the network learns using the other Layer 2 heads). When we fix the induction strength to -1, meaning the head is *only* paying attention to the wrong token, we recover an exponential curve. This is an example of the network using an "anti-induction" head (a phenomenon we sometimes observed if $L = 2$): the mechanism here is the network is *copying* both labels from context and then *subtracting out* the wrong label. This is an example of a slightly different (and perhaps more complicated) circuit for implementing ICL in our task, similar to how (Zhong et al., 2023) discovered two different circuits for modular addition. We hypothesize that the reason this "anti-induction" circuit doesn't emerge naturally when $L > 2$ is because imperfect versions of "anti-induction" heads (e.g., "Clamped IH strength" $= -0.8$) are less robust to different label pairs (middle plot, light-dark blue curves), which may make it a circuit element that is feasible but not learnable (under this task setup).

H. The "residual heads" hypothesis

In Section 3.1, we provide evidence that induction heads function additively with redundancy across heads. Combining this insight with the ordering of head emergence in Figure 3b, we hypothesize that other induction heads form to minimize the residual of the loss left after earlier heads form (e.g. Head 3). We didn't include this hypothesis in the main paper as it's not the focus of our work, and we find evidence supporting and opposing it. As scientists, we hope these mixed results can serve as a launch point for further study, and thus choose to include them here.

One simple way to test this hypothesis would be to consider the mistakes made by the individual Layer 2 heads, when all other Layer 2 heads are turned off. Let's consider Head 2 and Head 3 for example. Head 3 on its own achieves an accuracy of 98%, outputting the wrong answer for 1553 of the 78400 training sequences. Head 2's accuracy is 75%. If Head 2 is fitting to the residual, we might expect Head 2's accuracy on the sequences that Head 3 fails to solve to be above its baseline of 75%, meaning it's preferentially stronger on the problems where Head 3 is mistaken. On the other hand, it could be that the sequences missed by Head 3 are simply "harder", in which case we would expect an accuracy less than its baseline. In Table 2, we find that the latter is more likely to be the case, so this piece of evidence is *against* the residual heads hypothesis.

Table 2. Accuracy when ablating all but a given Layer 2 head on the subset of problems that are incorrect when ablating all but Layer 2 Head 3 (the one with highest induction score). Second row contains the accuracies over the whole dataset for comparison (which matches the circles in Figure 4a).

Accuracy	Head 0	Head 1	Head 2	Head 4	Head 5	Head 6	Head 7
On the subset	24.08%	64.78%	71.67%	32.97%	56.41%	16.61%	46.10%
Overall	40.32%	76.51%	75.12%	42.85%	64.65%	29.16%	58.85%

Next, we present some evidence that supports the hypothesis. Namely, we consider perfecting each induction head's match circuit (by setting its attention pattern to the perfect pattern that gives attention 1 to the correct token and 0 to all other tokens), which allows us to quantify the quality of its copy circuit. Then, we can examine the mistake rate of the copy circuit for different labels, and see these values across heads. Our results are summarized in Figure 16. Specifically, we do find some specialization of output classes: Head 3 has a higher mistake rate for copying labels 1 and 3, while heads 1 and 5 have virtually 0 mistake rate on these, indicating that they may have specialized to these examples. This result *supports* the "residual heads" hypothesis.

If we repeat the analysis of Table 2, we find some promising evidence in this "perfected match" condition (Table 3). These results suggest that, when it comes to the copying subcircuit, induction heads forming later in training preferentially focus on copying labels that earlier formed induction heads are worse at. As a result, these earlier heads do not have to "fix their mistakes" as much as if they were trained on their own, connecting to our results in Section 3.2. This result is *inconclusive* with respect to the "residual heads" hypothesis, as it only pertains to Head 1, and not the other heads.

Finally, we build off the result in Figure 4b, and try to take the argument further. Given that Head 3 is the quickest to learn and does in fact emerge first in the full network (Figure 3b), we consider training networks where all but *two* heads are ablated: Head 3 and each of the other 7 heads (one at a time). Loss curves are shown in Figure 17. By training with Head 3 and just one other head, we see loss curves are way closer to the actual training curve. However, the ordering of phase changes still does not match the ordering of emergence and strengths in Figure 3b, perhaps due to higher order interactions (e.g., between three heads or more). Given the current results though, this piece of evidence is *against* the "residual heads" hypothesis.

Given all these pieces of evidence, for and against, we cannot convincingly argue for or against the "residual heads" hypothesis. We hope this investigation (including negative results) can inform future work researching this hypothesis.

I. Reproducibility across model initializations

Our code, as well as notebooks reproducing findings on multiple initialization seeds (5 seeds) can be found at `https://github.com/aadityasingh/icl-dynamics`. We found little to no differences when training seed (responsible for data ordering) was varied. Here, we summarize qualitatively the results that differ slightly across init seeds and the ones

Table 3. Same as Table 2, except we also "perfect" each induction head's match circuit (as done in Figure 16), so we can view this table as only considering the mistakes in each induction head's copy circuit. There are only 58 examples that the "perfected match" layer 2 Head 3 gets wrong, so this table only considers the accuracy of other "perfected match" heads on these 58 examples. Note the overall accuracies in this Table match the squares in Figure 16.

Accuracy	Head 0	**Head 1**	Head 2	Head 4	Head 5	Head 6	Head 7
On the subset	24.14%	**86.21%**	20.69%	22.41%	50.00%	0.00%	44.83%
Overall	50.01%	82.80%	82.73%	62.10%	65.42%	28.46%	70.70%